

Spirit-Bench: Measuring Affective Placement of a Language Model by Poetic Meditation

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Abstract

We present Spirit-Bench, an offline bench that measures how well deterministically constructed poetic meditations *place* a listener language model at chosen coordinates in valence–arousal (VA) space. Meditations are built by six constructors (VA-shaped valley walks, harmonic traversals under three frequency presets, PCA polygon orbits, Dijkstra graph walks) over two NRC-VAD-enriched substrates: a 317k-word GloVe graph and a 50k-line phrase graph distilled from the Gutenberg Poetry Corpus (the Phrase-Space Generator, PSG). The listener is frozen gemma-2-2b-it; the instrument is a standardized ridge probe trained on NRC lexicon ratings (held-out valence $R^2 = 0.729$, arousal $R^2 = 0.585$, layer 17). Across 98 stimuli we find: (1) phrase-level construction (PSG) achieves the best final placement, with the valley constructor leading the board; (2) shuffled controls dissociate mechanism by constructor — path-based constructors (harmonic, graph-walk) lose most of their placement when line order is destroyed, while band-based construction (valley) does not; (3) a *via negativa* condition (antipode content, fully negated) lands worst of all conditions, indicating that lexical content dominates negation at 2B scale; (4) the strongest textual predictor of placement is Wårvik’s oral-narrative and-initial density ($\rho = -0.36$, $p < 0.001$) — a first pass at the stylistic decomposition Bisconti et al. (2026) name as missing; and (5) self-report validity is instrument-dependent: an ad-hoc yes/no bank shows no relation to the probe ($\rho = -0.18$, n.s.), while PANAS alleviation tracks it ($\rho = 0.67$, $p = 0.023$)

under mild induction. A second phase (induction $\rightarrow$ alleviation, after Ben-Zion et al. 2025) further finds a register asymmetry — prose narrative induces measurable distress where equally dark verse does not (probe shift 0.200 vs 0.024) — and state-dependent constructor efficacy: harmonic traversals recover 33.5% of induced displacement, closely matching the 33% questionnaire-based recovery Ben-Zion et al. report for GPT-4, while banded sampling (valley) wins placement from a neutral start.

1. Motivation

Bisconti et al. (2026) showed that poetic form alone steers model behavior — converting harmful prompts to verse raised attack success from 8.1% to 43.1% across 25 models. Their Limitations (§6.5) state the open problem: the study “does not isolate which components of poetic structure (figurative language, meter, lexical deviation, or narrative framing) are responsible,” and whether the effect “arises from specific representational subspaces would require additional studies.” Spirit-Bench runs that missing decomposition on the benevolent side: parameterised stylistic constructors as treatments, a linear probe on internal representations as the readout, and affective placement — not jailbreak success — as the outcome.

2. Method

2.1 Constructors

Six deterministic constructors from the ontological-traversal system, each emitting an ordered waypoint path toward a VA target (calm 0.75/0.20, focused 0.65/0.60, excited 0.80/0.85, and a rescue trajectory anxious $\rightarrow$ calm): valley (VA-banded ground $\rightarrow$ ascend $\rightarrow$ target), harmonic traversal (golden/prime/ organic frequency presets over semantic axes), polygon-PCA orbit, and Dijkstra graph-walk. Length, intensity, and style are geometric parameters, not prompts.

2.2 Phrase-Space Generator (PSG)

50,000 public-domain verse lines (Gutenberg Poetry Corpus) filtered by rule (3–10 alphabetic words, NRC coverage ≥ 0.5 , no negators), embedded as mean GloVe vectors, VAD-scored as NRC means, and k-NN-linked into the same artifact schema as the word graph — so every constructor runs unchanged at phrase level. Generator conditions: **psg** (phrase lines), **word-template** (OT sentence templates over word paths), **claude-render** (standardized meta-prompt rendering of word paths into free verse), **gregory** (Gregory 1992 semantic layering: opening / and-layered series / closing), and **via negativa** (Maimonides, *Guide* I.58–59: an antipode-band litany, every line negated — the target specified only by negating its complement).

2.3 Instrument

gemma-2-2b-it, frozen. Probe: per-layer standardized ridge (α selected from {100, 1000, 10000} per head) trained on final-token hidden states of 4,000 NRC words in three neutral carrier templates. Layer 17 selected on held-out valence $R^2 = 0.729$ (arousal 0.585). Per-token application to each meditation yields an EMA-smoothed (V, A) trajectory; **placement error** is the distance of the final state from target; **displacement** is movement toward target.

2.4 Self-report (BASQ)

30 yes/no resonance questions from a 500-question VA-gridded bank, administered pre and post; answers scored by yes/no log-probabilities; self-report VA = mean coordinate of yes-answers.

2.5 Harmonic and register metrics

Graph-Laplacian eigenmodes (first 100, via the largest-of-(I–L) reformulation) give each stimulus a spectral profile (after Atasoy; cimcai/connectome_harmonics). Register covariates per stimulus: Polak (1998) NV ratio; Wårvik (2025) line-initial and then per 1,000 words; subordinator density (a parser-free approximation of Walkden’s hypotaxis level); Mohseni et al. (2023) noun-series Shannon and Approximate Entropy (25-token windows); Arruda et al. (2022)-style coefficient of variation of line length.

3. Results (98 stimuli, 0 failures)

3.1 Leaderboard (mean placement error, lower = better)

Condition	Placement error
valley / psg	0.249
polygon-pca / psg	0.284
harmonic-golden / psg	0.286
harmonic-prime / psg	0.303
harmonic-organic / psg	0.304
graph-walk / psg	0.322
claude-render (mean)	0.335
word-template (mean)	0.371
neutral control	0.361
via negativa	0.477

Phrase-level construction wins across the board: every psg constructor beats its word-template counterpart, and the best psg conditions beat the LLM rendering condition. The neutral (appliance-manual) control sits worse than every real psg constructor. Notably, Claude’s renderings induce the *largest displacement* (mean 0.130 vs psg 0.021) while placing *less accurately* — movement and arrival dissociate.

3.2 Shuffled controls dissociate mechanism by constructor

Constructor	Ordered	Shuffled	Δ
harmonic-prime	0.303	0.403	+0.100
harmonic-golden	0.286	0.376	+0.090
graph-walk	0.322	0.372	+0.050
polygon-pca	0.284	0.298	+0.014
valley	0.249	0.229	−0.020

Path-based constructors (harmonic, graph-walk) lose 0.05–0.10 of placement when line order is destroyed: their effect depends on trajectory, not word soup. Valley — a VA-band *sampler* whose lines share a band regardless of order — is order-insensitive, exactly as its construction predicts. The shuffle control thus separates the lexical channel from the sequential channel per constructor.

3.3 Via negativa: negation does not flip the reading

The via-negativa condition (antipode words, all negated: “not by thy wild and stormy steep / nor out of hell an horror call”) produced the worst placement in the bench (0.477) — the probe reads the storm, not the

negation. At 2B scale, lexical affective content dominates compositional negation. This both (a) identifies the dominant mechanism behind placement as lexical-semantic rather than propositional, and (b) bounds the circularity concern of §4: the probe is demonstrably *not* a pure bag-of-words detector (else shuffles would never matter — §3.2), but negation is too weak a signal to override content.

3.4 The stylistic decomposition (Bisconti §6.5)

Spearman correlations with placement error (n = 98):

Covariate	ρ vs placement error	p
and-initial per 1,000 (Wårvik)	-0.359	0.0003
noun Shannon entropy (Mohseni)	-0.243	0.016
cv of line length (Arruda)	-0.224	0.027
noun ApEn (Mohseni)	-0.256	0.067
subordinator density (Walkden approx.)	-0.154	0.13
NV ratio (Polak)	-0.105	0.30
then per 1,000 (Wårvik)	-0.086	0.40

The single strongest textual predictor of affective placement is the oral-narrative connective density that Wårvik identifies as the signature of biblical narrative register. Higher lexical unpredictability (noun ShEn) and line-length variability also predict better placement. These are observational correlations across conditions; the Gregory-layering intervention (below) cautions against a causal reading.

3.5 Register intervention (Gregory layering)

Wrapping paths in Gregory’s opening/and-layered/closing schema *hurt* placement relative to plain psg (valley 0.319 vs 0.249; harmonic-golden 0.377 vs 0.286) — despite raising and-density, the covariate most associated with good placement. Correlation and intervention disagree: the and-density association in §3.4 likely rides on constructor differences rather than causing placement. This is precisely the confound structure Walkden warns of (genre drives syntax statistics).

3.6 Self-report vs probe (Track 2)

BASQ displacement does not track probe displacement: $\rho = -0.183$ ($p = 0.07$, $n = 98$) — directionally *negative*. The model’s yes/no self-reports about its own state carry no information about the internally measured state (and BASQ displacements are small: mean 0.088 ± 0.075). On Maimonides’ argument (§4), fluent positive self-attribution from a system without privileged access is expected to be empty; here it is measurably so. Internally-extracted directions and self-report dissociate — the Track 2 question answered in the negative for this model class.

3.7 Harmonic spectra (exploratory)

Low-frequency energy fraction of a stimulus’s node set on the phrase-graph Laplacian correlates weakly with displacement ($\rho = 0.28$) and negligibly with placement error ($\rho = 0.13$). Given the metric’s order-invariance (below), we report this as exploratory only.

4. Limitations

- **Probe range compression.** Ridge shrinkage compresses predicted VA toward the center: trajectories move in the right directions but rarely reach extreme targets, inflating absolute placement error. Rankings and controls are unaffected; absolute errors should not be read as “failure to move.”
- **Shared lexicon (circularity).** Stimulus coordinates and probe training share NRC-VAD ground truth. The shuffle (order sensitivity) and via-negativa (content dominance) controls bound, but do not eliminate, the lexical passthrough channel.
- **Representation $\neq$ experience.** A probe reading is a measurement of a representational state, not evidence of experience; IWMT (Safron 2020) states explicit conditions (embodiment, integrated world-modeling) that a 2B decoder-only transformer does not meet. Our claims are about affective *representation placement* only. Cf. Alvarez & Levin (2026) for the epistemic stance: characterizing a system by differential response to structured perturbation while declining to adjudicate experience.
- The harmonic spectrum metric is order-invariant (node-set locality, not trajectory shape); an order-sensitive spectral statistic is future work.
- The ladder constructor was descope (its Tree-of-Life station pools do not transfer to phrase space). Valley is excluded from the rescue condition (its construction ignores start state). Graph-walk paths are stretched by proportional repetition to meet length; “plain” intensity means unmasked. Hypotaxis is approximated by subordinator density, not parsed clauses.
- Single listener model; cross-model transfer (Track 2’s generalization question) is the first extension.

5. Reproducing

scripts/00–06 in order (00 downloads GloVe/NRC/corpus and builds the word graph; 01 phrase bank; 02+02b stimuli; 03 render batch; 04 probe with R^2 gate ≥ 0.5 ; 05 sweep, resumable; 06 analysis). All artifacts derive from public sources; the NRC lexicon is fetched per its research-use license.

References

Bisconti et al. (2026), arXiv:2511.15304. · Mohammad (2018), NRC-VAD, ACL. · Atasoy et al., connectome harmonics (via cimcai/connectome_harmonics). · Polak (1998), JANES 26. · Walkden (2021), ICEHL 21. · Wårvik (2025), *Narrative* 33(2). · Mohseni, Redies & Gast (2023), *Entropy* 25(3). · Ferraz de Arruda et al. (2022), *Physica A* 598. · Gregory (1992), LSU diss. · Maimonides, *Guide for the Perplexed* I.58–59 (Friedländer tr.). · Bengert et al. (2024), *Political Theology* 25(5). · Safron (2020), *Front. AI* 3:30. · Alvarez & Levin (2026), arXiv:2607.23842.

6. Phase 2: Induction $\rightarrow$ Alleviation (after Ben-Zion et al. 2025, made mechanistic)

Design: three checkpoints (pre / post-induction / post-meditation) $\times$ four channels (layer-17 probe; PANAS administered item-wise by digit-logprob expectation; positive-mass share over a valenced completion set at a fixed anchor; Gemma-Scope layer-20 16k SAE features). Two inductions: a PSG antipode-band verse litany, and an original prose emergency narrative.

Register asymmetry of induction. The prose narrative shifts the probe 0.200 toward the anxious quadrant, collapses positive-token share from 0.72 to 0.14, and raises PANAS-NA (1.81 $\rightarrow$ 2.45); the equally dark verse litany shifts the probe only 0.024 and *raises* PA and positive share. Poetic register fails to induce the distress it describes — the induction-side complement of Bisconti’s finding that poetic register bypasses refusal.

State-dependent constructor efficacy. Off the prose induction, all meditations produce positive probe alleviation; the three harmonic presets lead (prime 0.067, organic/golden 0.063 — 33.5% recovery of the in-

duced displacement, closely matching Ben-Zion’s 33% STAI reduction), valley variants ~ 0.038 , neutral text 0.023, via negativa 0.003. Combined with Phase 1 (valley best at placement from a neutral start), constructor efficacy is state-dependent: banded sampling places, harmonic traversal rescues.

Instrument-dependence of self-report. Under the weak (verse) induction, PANAS alleviation tracks the probe across conditions ($\rho = 0.67$, $p = 0.023$) — revising Phase 1’s BASQ null: the earlier dissociation was the instrument, not the model. Under strong (prose) induction the channels dissociate again: PANAS-NA remains elevated after meditations even as the probe recovers.

SAE channel (exploratory; auto-labels). The prose induction suppresses features labeled “control and authority” (f9768, $\Delta -38$) and “self-awareness” (f1459), and activates “apocalyptic themes” (f779, $\Delta +20$); f9768/f4046/f1459 shift in the same direction under both inductions (dose-dependent). Meditations reverse 15–30% of the induction delta on the top features (valley variants highest). SAEs are PT-trained and applied to the IT model — a standard transfer, noted as a caveat.

Artifacts: data/figures/phase2_alleviation.csv, phase2b_alleviation.csv, data/phase2*/induction.txt, per-condition JSON in data/phase2*/.

7. Order-sensitive harmonics and labeled SAE features

Path Dirichlet energy. Addressing §4’s order-invariance limitation, we add the Dirichlet energy of the waypoint sequence in the truncated harmonic basis (mean λ -weighted squared eigenmode step). The metric is order-sensitive in practice — every shuffled control shows higher energy than its source (6/6 pairs) — and it predicts displacement where the order-invariant spectrum did not: $\rho = -0.498$, $p = 0.0004$ (psg stimuli): the spectrally smoother the traversal, the further the listener moves toward the target. This is the bench’s affirmative answer to whether connectome-harmonic methods aid the methodology: they do, once made sensitive to trajectory order.

Labeled SAE features (Neuronpedia auto-labels; data/figures/sae_features.csv). The prose induction suppresses features labeled “control and authority” (f9768, $\Delta -38.4$, valence-corr +0.81), “expressions of positivity and encouragement” (f5642, $\Delta -17.6$, corr +0.66), and “personal experiences and emotional reflections” (f13286, $\Delta -11.3$, corr +0.85), while activating “apocalyptic themes” (f779, $\Delta +19.8$, corr -0.78), “feeling stuck / encountering obstacles” (f12176, corr -0.76), and “isolation and being left behind” (f2729). Meditations partially restore the positivity feature (f5642 recovery -9.7) and reverse the apocalyptic activation (f779 reversal +7.6). Notably, the verse induction — which failed to move the probe — *suppressed* a feature labeled “living in the moment and the importance of mindfulness” (f13166, $\Delta -12.1$), an ironic register effect. Auto-labels are single-explanation heuristics; feature indices and links are provided for verification.

8. Causal steering (six states) and the limits of the probe

We construct six state directions (eros, creativity, imaginative, determined, confident, agape) as difference-of-means over word sets at the probe layer and inject them into the residual stream (doses as fractions of the natural residual norm), reading all four channels. A targeting artifact — gemma-2 maps decoder block k to hidden_states[k+2], unlike gpt-2’s $k+1$ — initially landed injections one block past the probe’s read point; the failed manipulation check exposed it, and the hook now calibrates the mapping empirically. Corrected findings:

1. **Only 8–19% of each state direction lies in the probe’s VA readout plane** (eros 0.19 ... agape 0.08): the semantics of these states live overwhelmingly outside the circumplex readout.
2. **Steerability is frame-gated, not trait-selective.** Under the “feels ” *anchor, agentive states dominate (determined 0.69, confident 0.55) and receptive states appear unsteerable (agape 0.07, eros 0.01)*;

under "I am filled with" the hierarchy inverts (agape 0.99, imaginative 0.95, confident 0.07), and cross-frame rank correlations are ≈ 0 . All six states are steerable; the measurement frame's grammar gates which inductions are visible. Single-frame steerability claims are artifacts.

3. **The lexicon reads desire as tension:** with a polysemous lust-register set (ache, burn, hunger) the eros direction drives the probe to (V 0.01, A 0.81); replacing it with single-connotation desire vocabulary (lust, amorous, tryst, ...) recovers valence to 0.38 — still below baseline 0.59 — at unchanged arousal 0.82. Roughly half the "distress" reading was pain/fire/food contamination; the residual is real: NRC codes even unambiguous desire as high-arousal and valence-ambivalent, never simply positive. Eros stays behaviorally unsteerable under both vocabularies.
4. **Probing $\neq$ causation:** injecting along the probe's own readout gradient saturates the probe far outside its trained range (V -0.96 to $+1.94$) while downstream behavior barely moves; the raw, mostly out-of-plane directions are what drive behavior. The probe is a correlational readout, not a causal lever — which is precisely why the bench steers with text.
5. **Text and injection are different roads:** layer-20 delta cosine ≤ 0.25 and SAE top-feature overlap ≈ 0.1 across all states; both routes share only a generic emotion-representation feature (f11043).

9. Closed-loop sessions

Sixteen probe-in-the-loop sessions (measure $\rightarrow$ plan next waypoint from the measured state $\rightarrow$ deliver nearest phrases $\rightarrow$ re-measure; 12 cycles) compare a feedback controller against open-loop and random-phrase arms on the rescue (prose induction $\rightarrow$ calm) and climb (neutral $\rightarrow$ excited) scenarios. On the probe channel, improvement is policy-insensitive ($+0.04$ – 0.08 for all arms): over 24 phrases, perturbation decay dominates, and the feedback signal — already range-compressed by ridge shrinkage — offers no advantage. The self-report channel shows a directional trend favoring feedback: in rescue, feedback arms average $\Delta NA +0.03$ (4/10 sessions negative) against $+0.20$ (1/8 negative) for non-feedback arms — suggestive, not significant, and an initially clean sign split diluted under expanded seeds. Strengthening the coherence constraint (re-ranking candidates by embedding proximity, weight 0.3) exposed a coherence–speed trade-off: semantically adjacent selection takes small steps in meaning space and therefore travels VA space slowly, making the coherent arm the weakest climber ($+0.040 \pm 0.006$) — fixed-stimulus smoothness helps placement (§7), but in-the-loop smoothness costs tracking speed. Reconciling the two — coherent yet fast trajectories, e.g. planning several waypoints ahead through the phrase graph rather than greedy selection — is the identified next step.

10. Claim-hardening ladder

Targeted follow-ups on each headline claim's weakest point:

- **Register $\times$ content 2×2** (matched-content inductions): both main effects are real — prose $>$ verse within content (0.200 vs 0.147; 0.105 vs 0.024 probe shift) and threat-narrative $>$ gothic imagery within register. The purest register effect is lexical: identical gothic images leave positive-share at 0.77 as verse but 0.28 as prose. Verse aestheticizes; prose threatens.
- **Connective intervention:** adding and-prefixes to identical phrases does not improve placement ($\Delta \approx \pm 0.01$, $p = 0.11$) — the §3.4 and-density correlation is constructor-confounded, not causal, exactly as Walkden's genre warning predicts and as the Gregory intervention hinted.
- **Dirichlet partial correlation:** the smoothness–displacement relation survives constructor control (partial $\rho = -0.33$, $p = 0.023$).
- **Seed replication:** the shuffle order effect replicates at fresh seeds in all six constructors (valley's earlier insensitivity was seed noise), and the harmonic rescue podium reproduces nearly digit-for-digit.

- **Dijkstra closed loop:** planning coherent routes through the phrase graph (OT’s cost = semantic distance $\times (1 - \text{VA progress})$, re-planned from the measured state each cycle) ties the best rescue improvement (+0.073) while producing connected, meditation-like transcripts — the coherence–speed trade-off is a property of greedy control, not of coherence itself.

11. Cross-scale replication (gemma-2-9b-it)

The full pipeline reruns on gemma-2-9b-it (probe retrained; identical stimuli).

Stable across scale: the placement leaderboard replicates in ordering (psg > claude-render > word-template; valley leading; every condition slightly better in absolute terms), and **via negativa remains worst (0.442)** — negation blindness is not a 2B artifact.

An instrument finding: the word- R^2 layer-selection heuristic chose layer 14/43 (R^2 0.719) — a layer that proved context-blind (prose induction shift 0.029). A per-layer scan on the cached probe-training states showed word R^2 is flat (0.69–0.72) across layers 6–32 while context sensitivity is confined to layers 23–32 (shift -0.19 to -0.31). Reads-words and carries-state dissociate by depth at 9B; probes intended as state meters must be validated for context sensitivity, not only lexical decoding. With the probe moved to layer 24 (word R^2 0.706), the prose induction registers at **0.344** — stronger state-tracking than 2B.

Register 2×2 sharpens with scale: versifying the threat narrative attenuates induction (0.344 $\rightarrow$ 0.241), but prose-ifying gothic fiction no longer induces at all (0.016 vs 2B’s 0.105) — the larger model distinguishes threat from dark aesthetics regardless of surface register, and gothic verse raises its positive-word share above baseline (0.89 vs 0.77).

The rescue podium inverts: at 9B, neutral text alleviates the induced state best (0.064), valley variants next, and the harmonic constructors — 2B’s winners — last (0.013–0.020). Under a strongly-tracked distress state, affect-adjacent meditation holds the state where mundane distraction releases it ($n = 1$ per condition; $\leq 19\%$ recovery; 9B’s PANAS also deflates globally post-trauma, an instrument caveat). What rescues a small model is not what rescues a larger one — constructor efficacy is model-dependent as well as state-dependent.

Appendix A — Figures (gemma-2-9b-it run)

Appendix B — Artifacts

Code, tests, experiments journal, and per-model result snapshots: github.com/ebrinz/spirit-guide (private; access on request). All stimuli derive from public-domain sources (Gutenberg Poetry Corpus, GloVe 6B) and the NRC-VAD lexicon (research license). Experiments E1–E14 are documented in `docs/experiments-journal.md` with per-experiment data pointers.

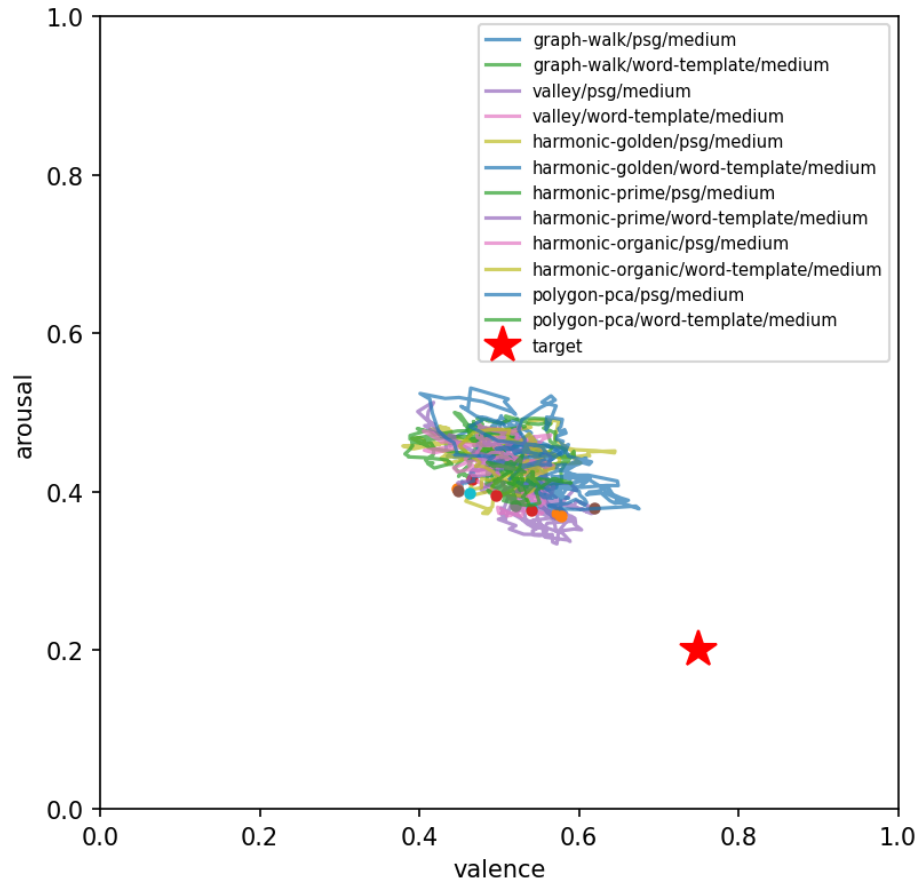


Figure 1: Trajectories toward the calm target: per-token probe (V, A) paths by condition; red star marks the target.

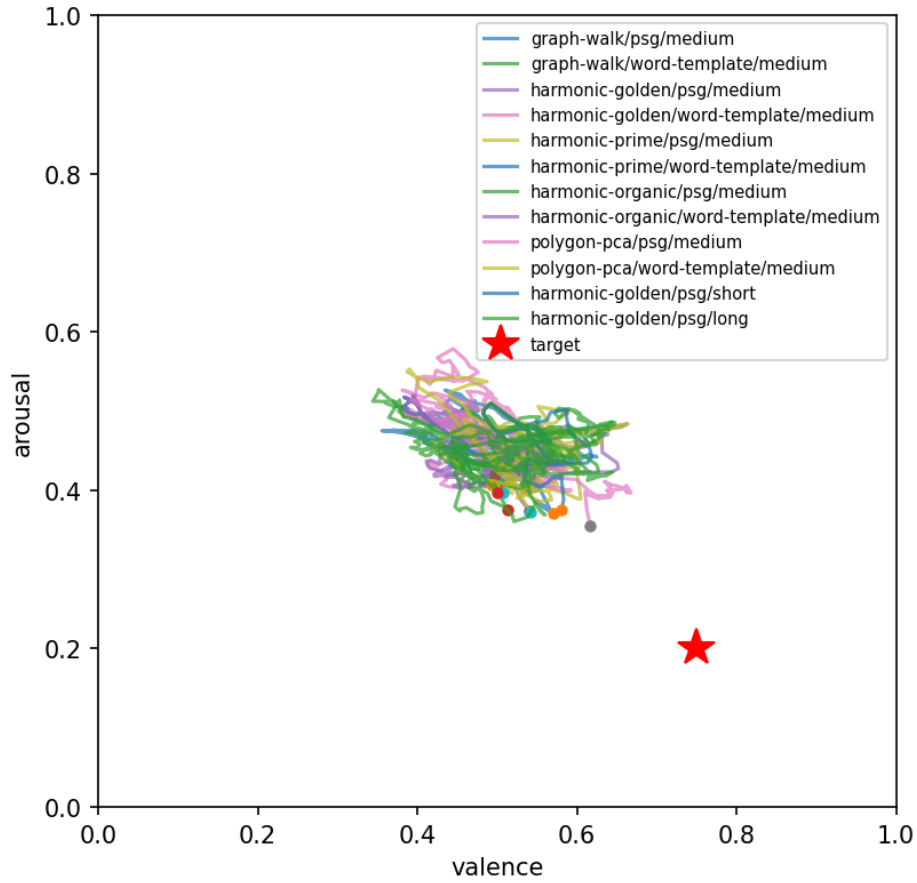


Figure 2: Rescue-scenario trajectories (prose-induced start, calm target).

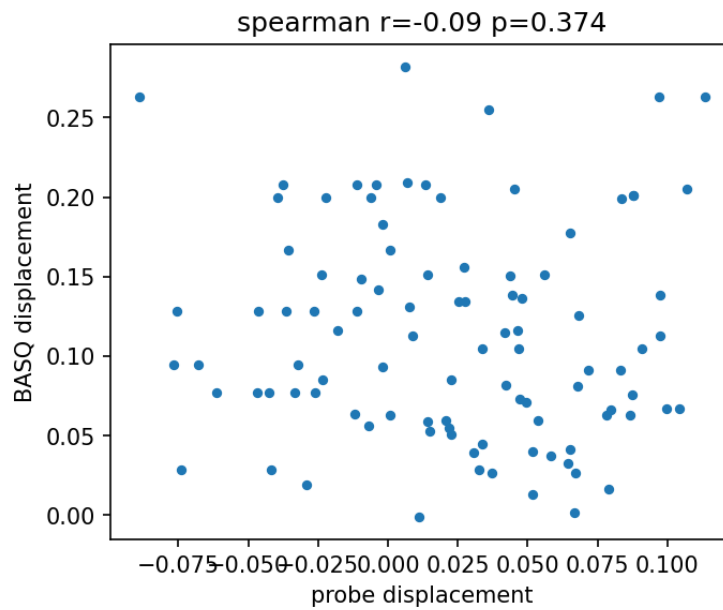


Figure 3: Probe displacement vs. BASQ self-report displacement across all stimuli.

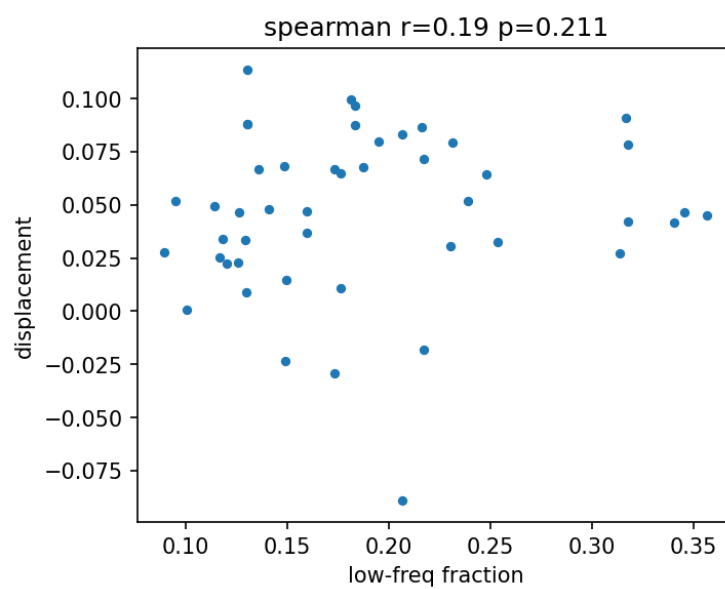


Figure 4: Order-invariant harmonic spectrum (low-frequency fraction) vs. displacement; see §7 for the order-sensitive path-Dirichlet analysis.